
Architecture of Database Management System

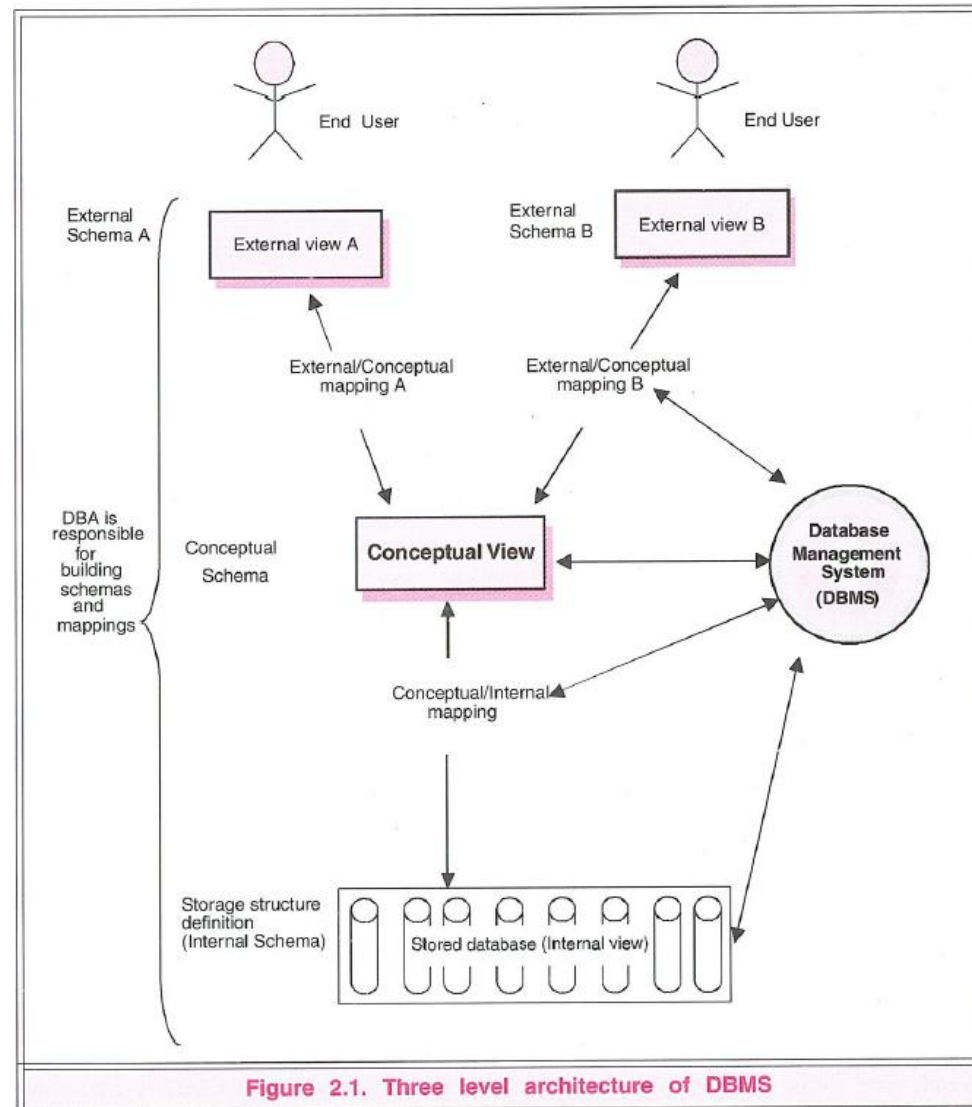
Early Two Level Architecture

- Given by Data Base Task Group (DBTG) in 1971.
 - Early attempt to standardize database development.
 - Recommend two level approach.
 - Schema
 - Subschema
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Objective of the Three Level Architecture

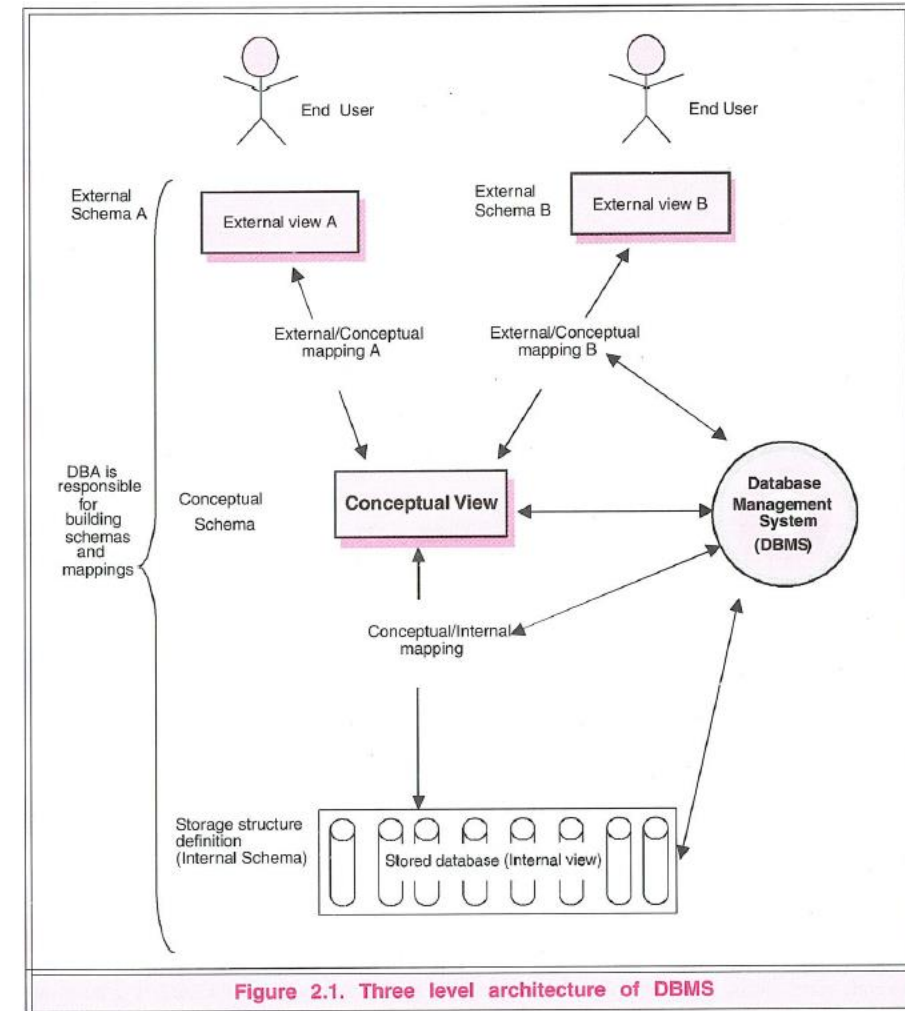
- Each user should be able to access the same data, but have a different customized view of the data.
 - User's interaction with the database should be independent of storage considerations.
 - The Database Administrator (DBA) should be able to change the database storage structures without affecting the user's views.
 - The internal structure of the database should be unaffected by changes to the physical aspects of storage, such as the changeover to a new storage device.
 - The DBA should be able to change the conceptual structure of the database without affecting all users.
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Three Level Architecture of DBMS



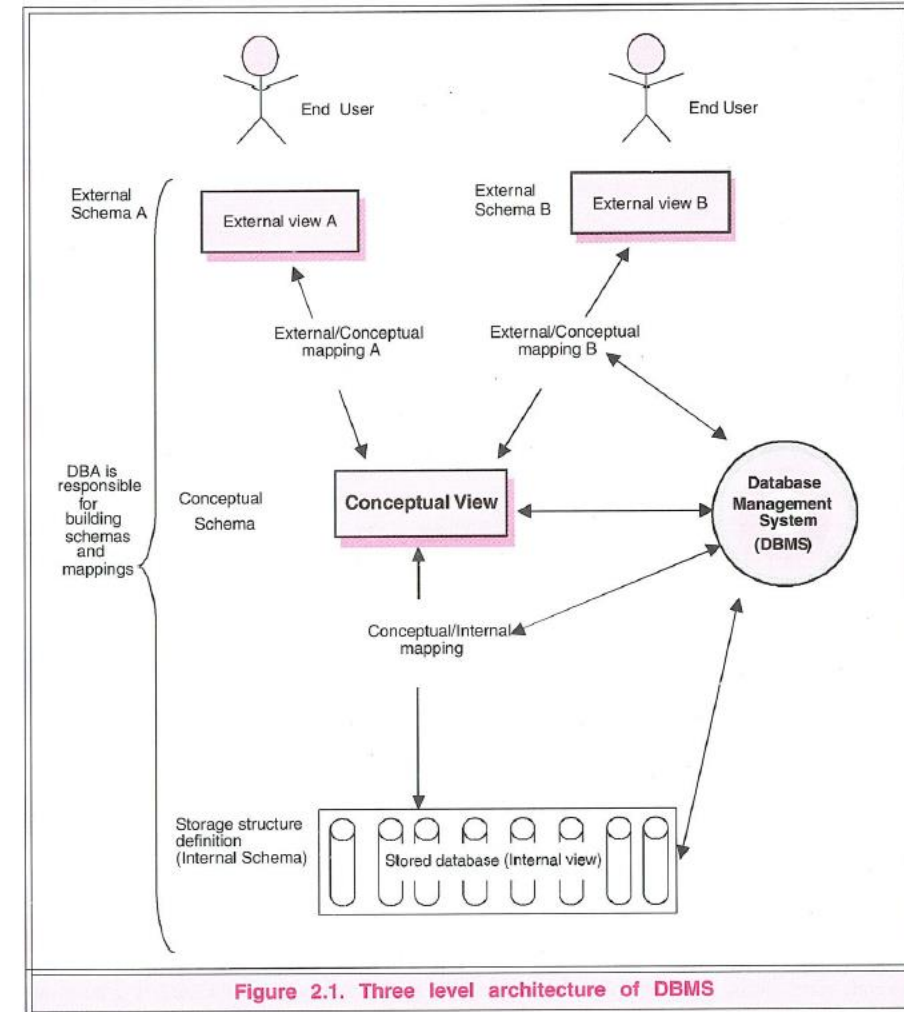
Three Level Architecture of DBMS

- External Level
- Conceptual Level
- Internal Level



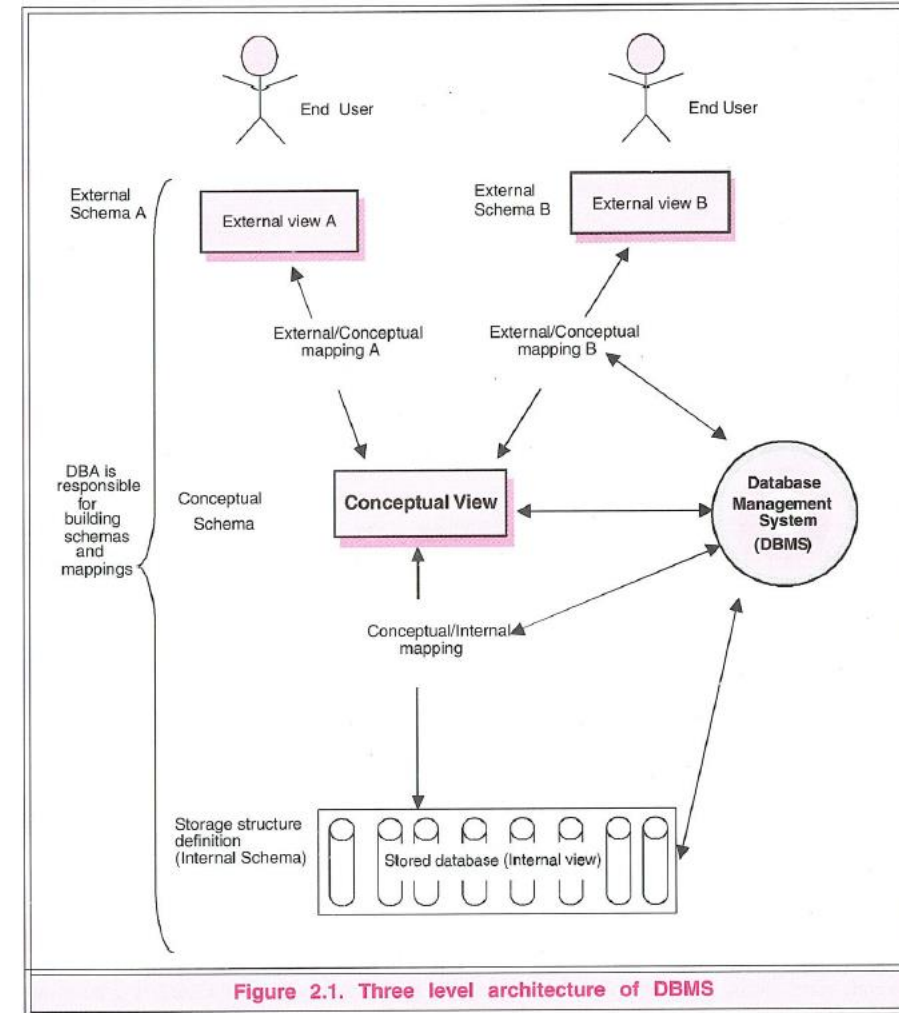
External Level or View level

- It is the users' view of the database.
- It describes that part of the database that is relevant to each user.
- It is closest to the end users.
- External level is also known as the view level.



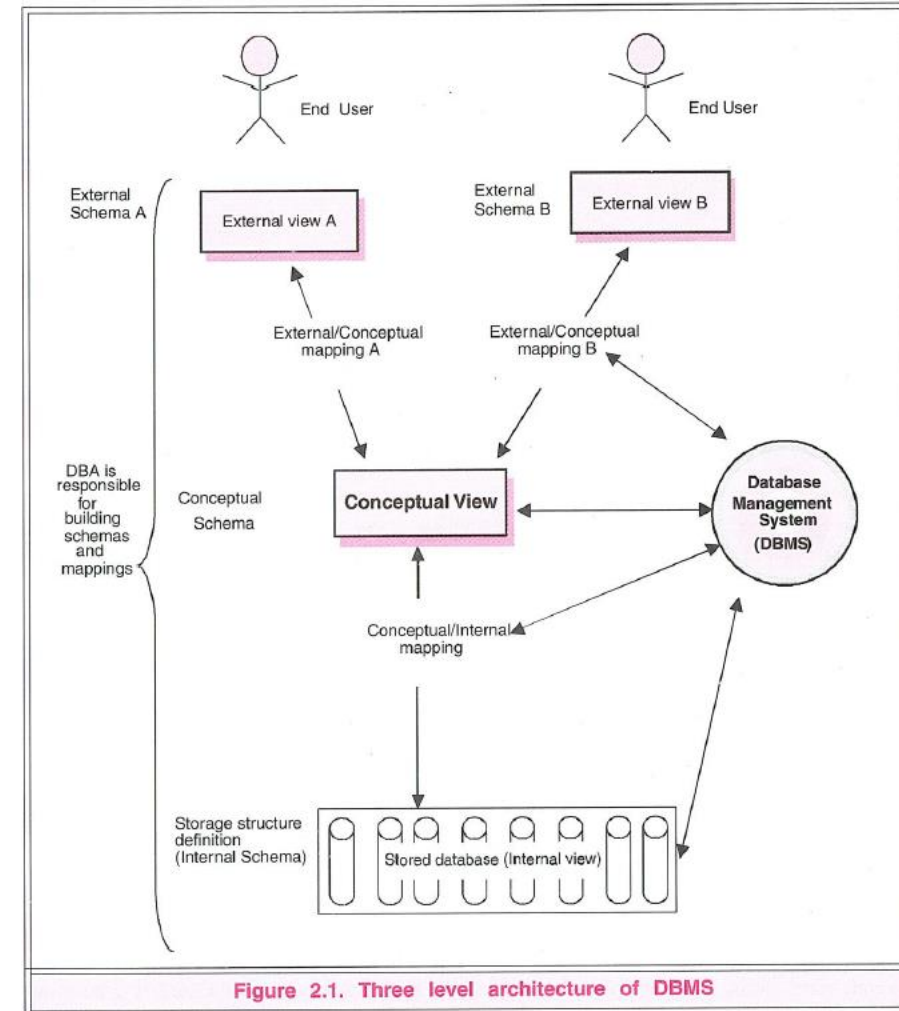
External Level or View level

- It deals with the way in which individual users view data.
- Individual users are given different views according to their requirements.
- Example: one user may view dates in the form (day, month, year), while another may view dates as (year, month, day).
- One may be interested in First Name other may be in First Name and Last Name.



Conceptual Level or Logical Level

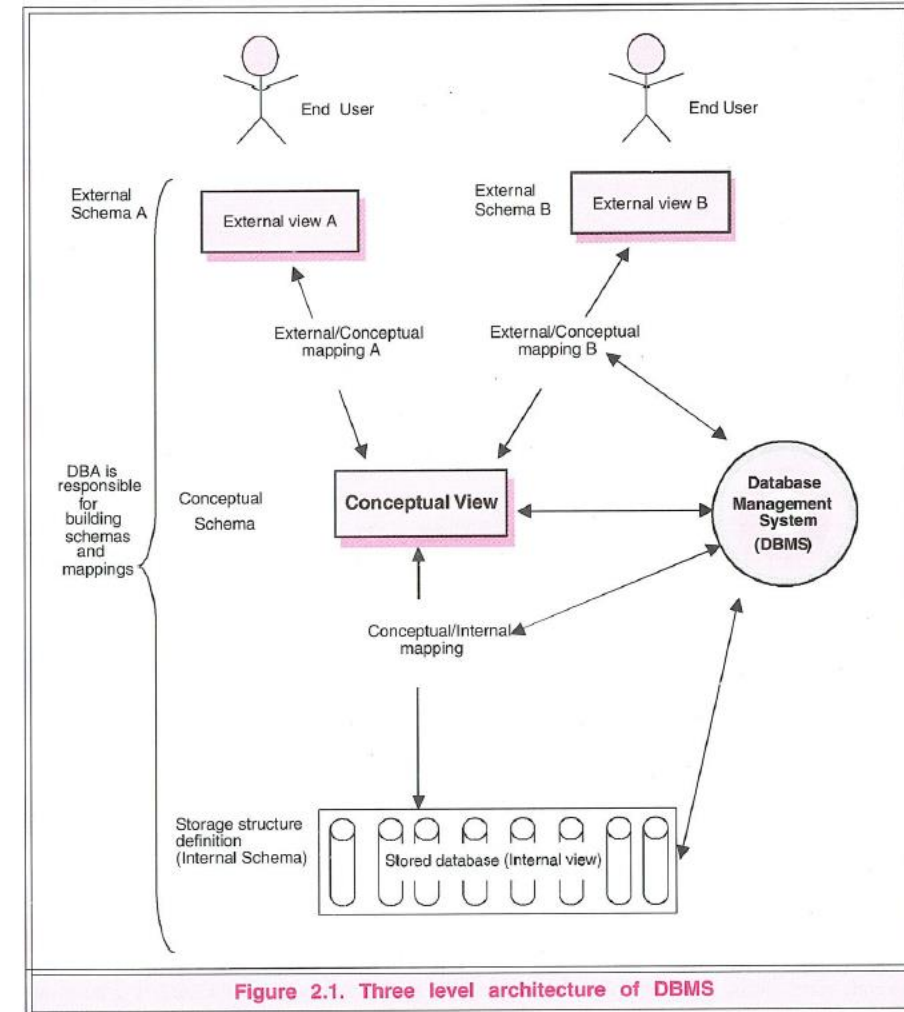
- This level must not contain any storage dependent details.
- For example it contains only information about data type and its size but not any storage considerations, such as the number of bytes occupied.
- Conceptual level is also known as the logical level.



Conceptual Level or Logical Level

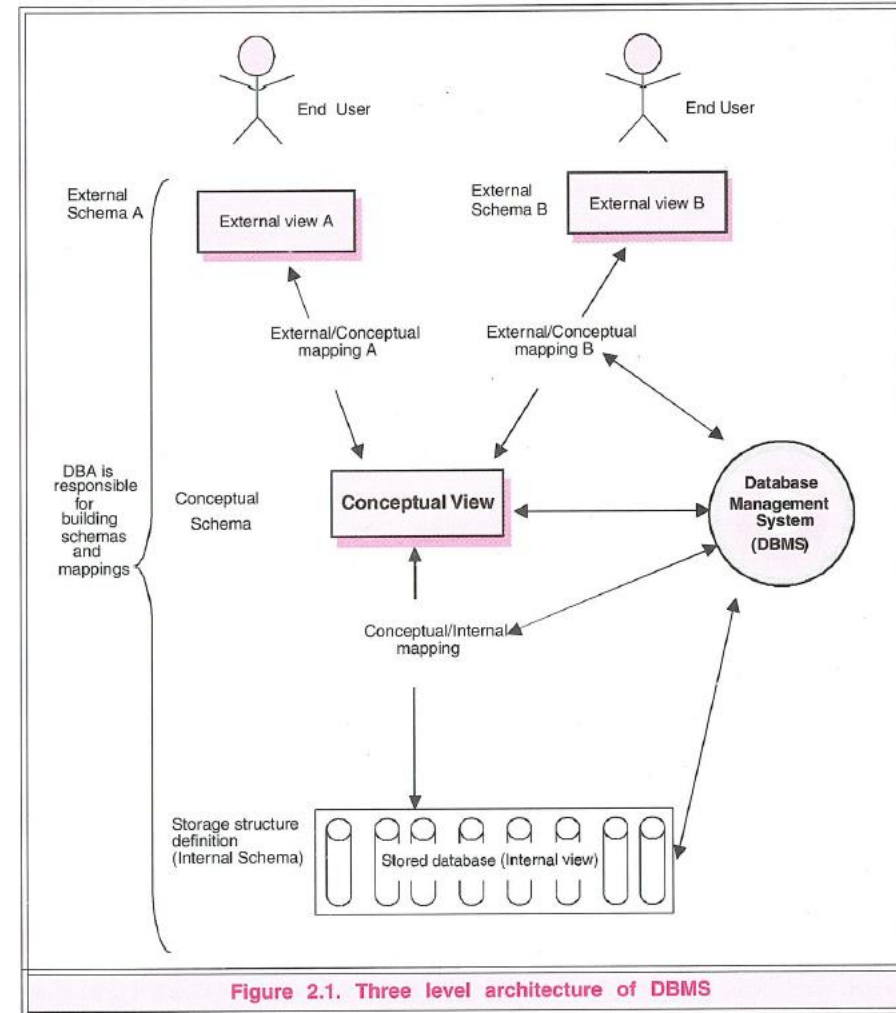
This level deals with

- All entities, their attributes, and their relationships
- The constraints on the data
- Security and integrity information



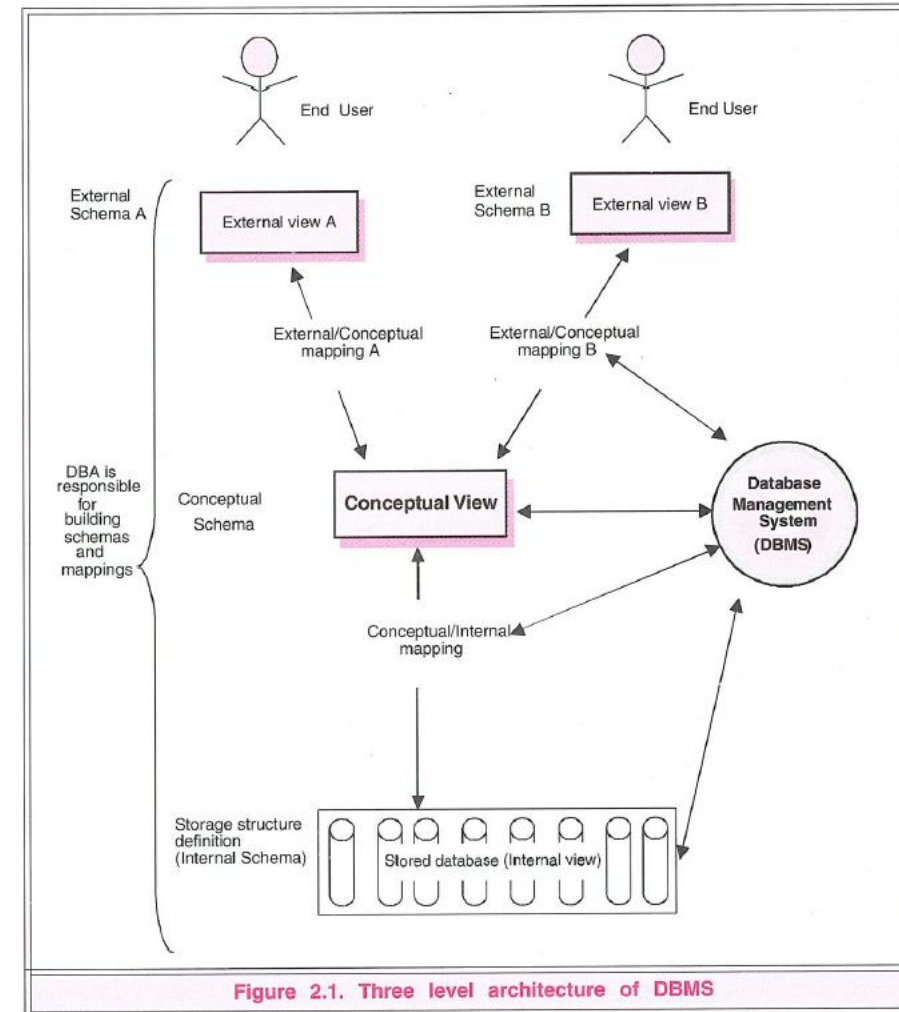
Internal Level or Storage level

- It is the physical representation of the database on the computer.
- This level describes how the data is stored in the database.
- It concerns the way the data are physically stored on the hardware.



External Level or View level

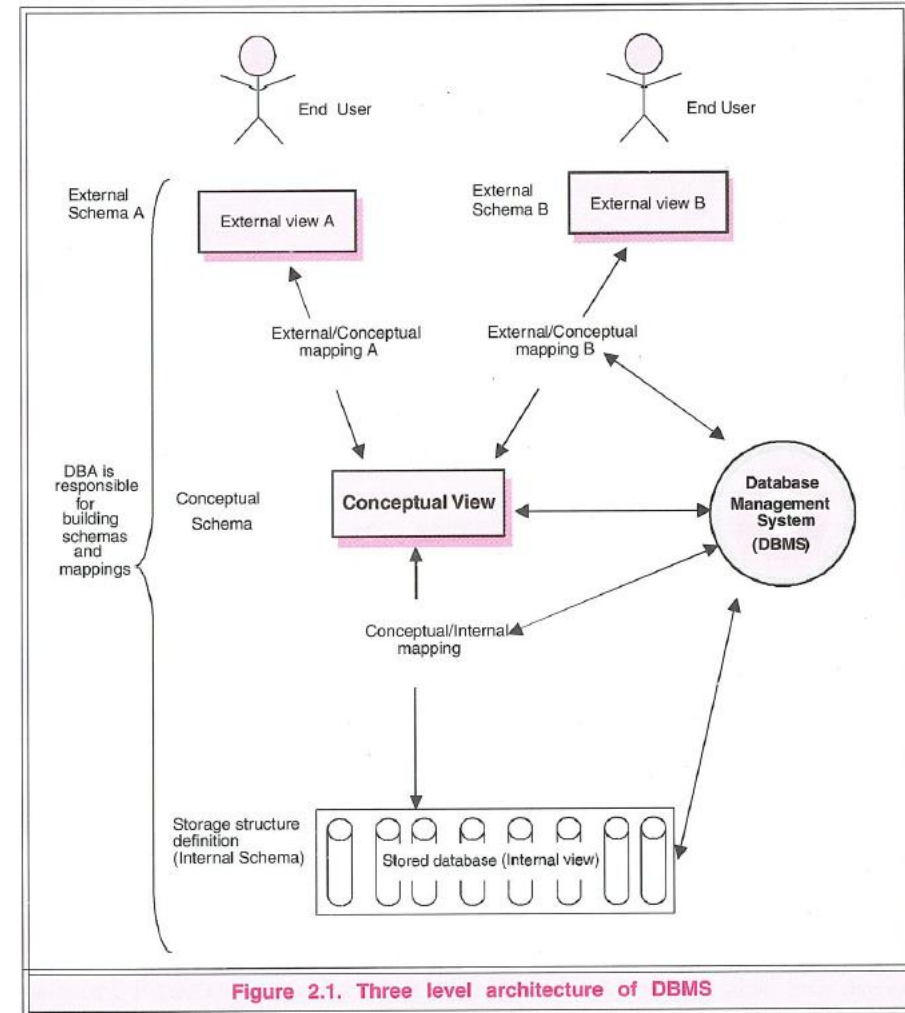
- The internal level covers the physical implementation of the database to achieve optimal runtime performance and storage space utilization.
- It covers the data structures and file organizations used to store data on storage devices.



Internal Level or Storage level

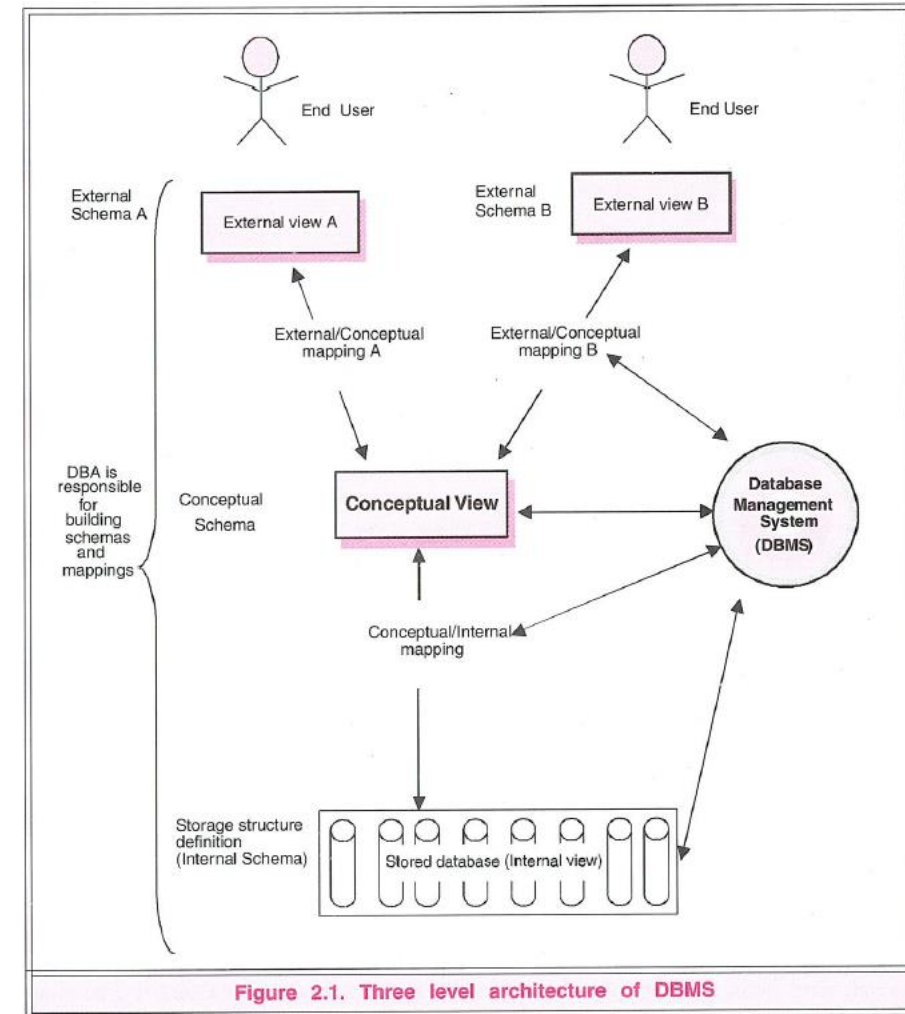
The internal level is concerned with

- Storage space allocation for data and indexes;
- Record descriptions for storage (with stored sizes for data items);
- Record placement;
- Data compression and data encryption techniques.

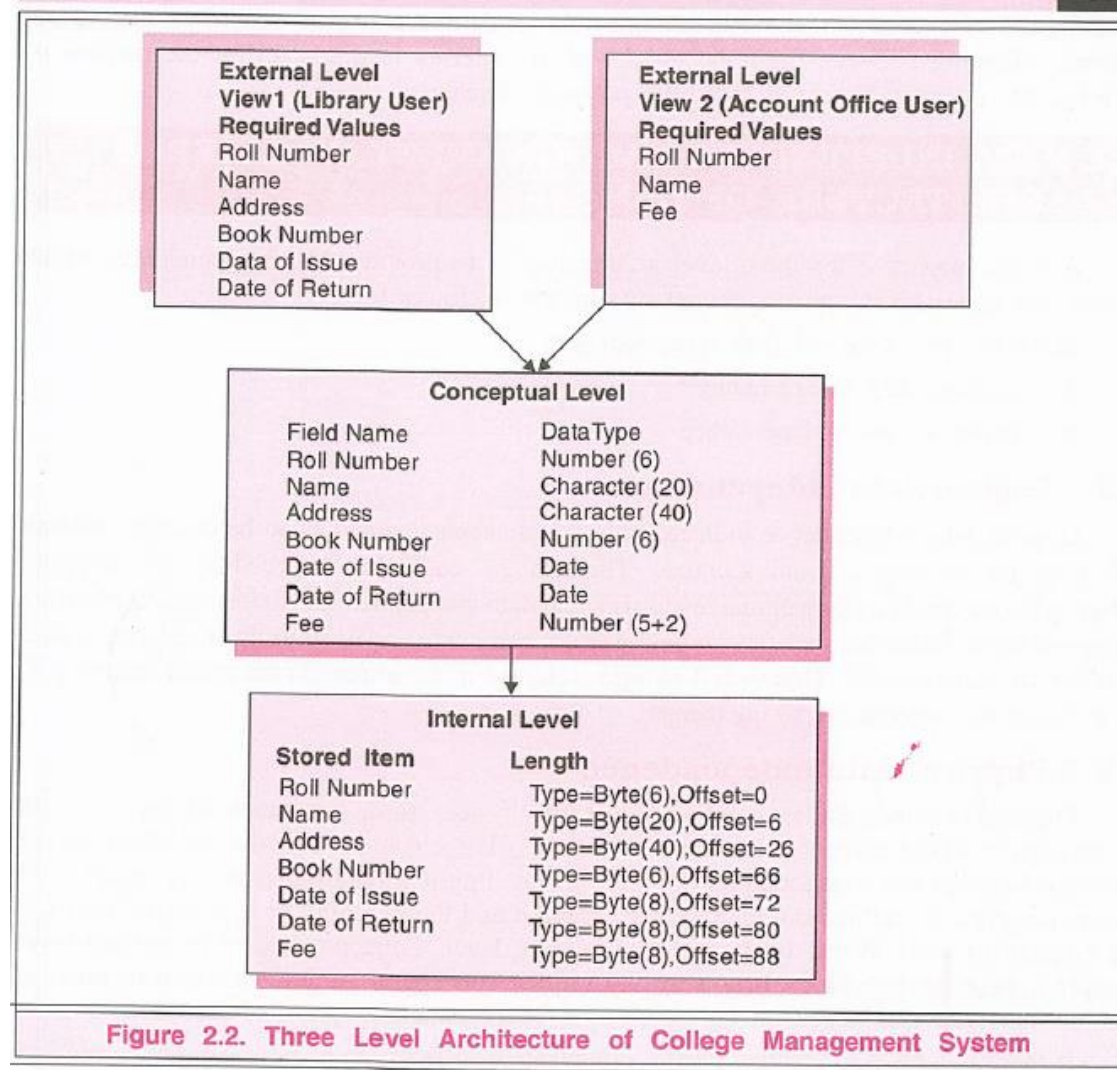


Three Level Architecture of DBMS

- There will be only one conceptual view.
- Similarly there will be only one internal or physical view, representing the total database, as it is physically stored.
- But there are multiple external views corresponds to each type of users.



Example: University Management System



Data Independence

Achievement of Three Level Architecture

Data Independence

- Application Programs should be independent from the way the data is stored and accessed.

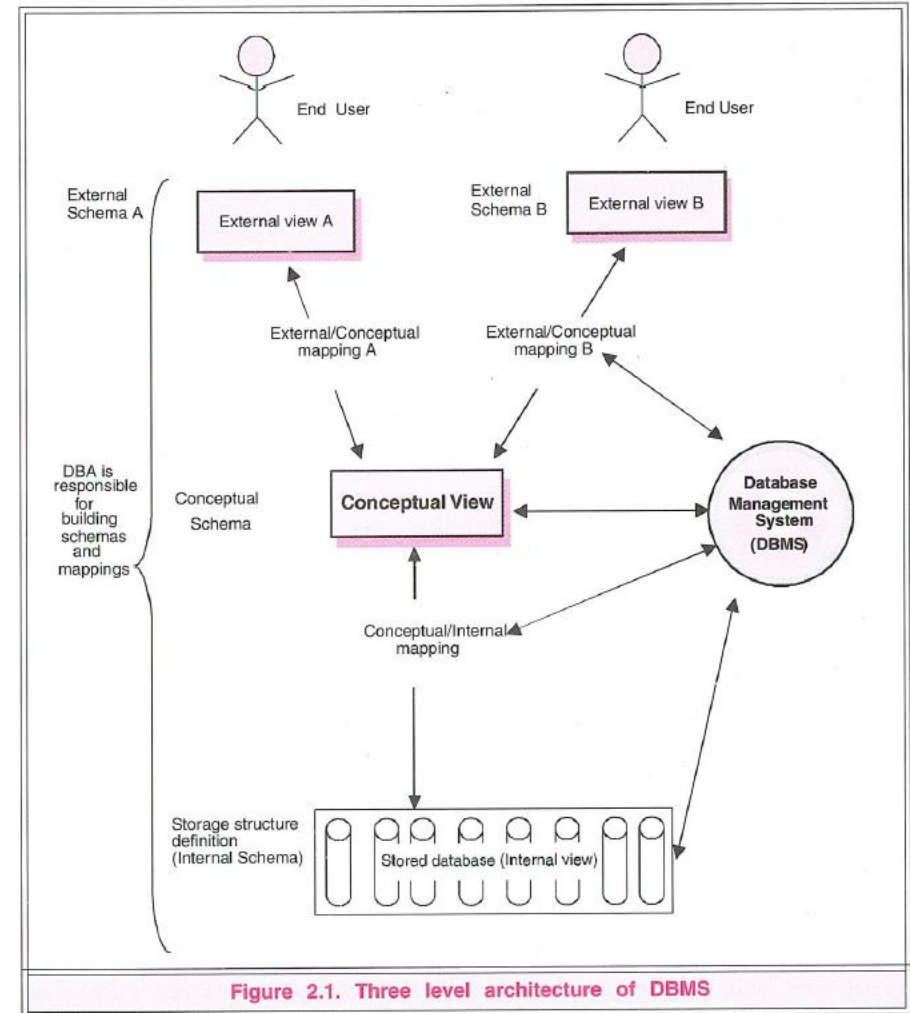
Data Independence-Achievement of Layered Architecture of DBMS

There are two kinds of data independence:

- Logical data independence
- Physical data independence

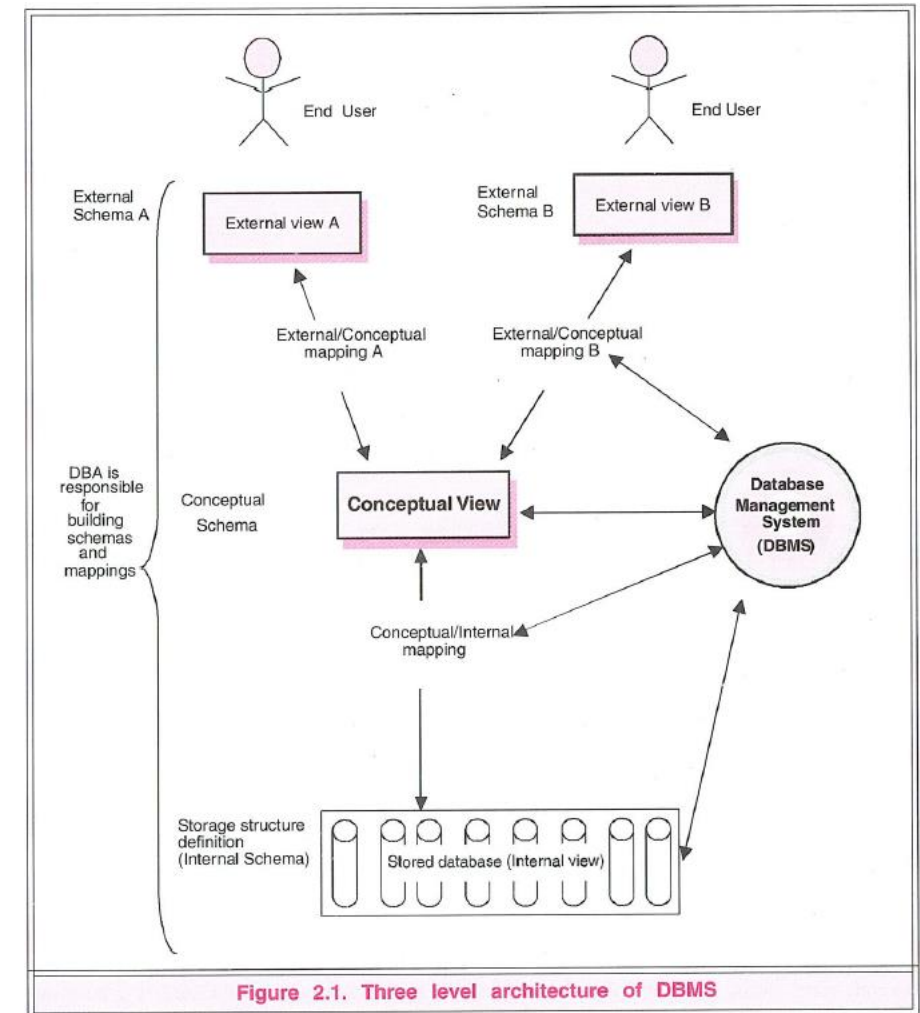
Logical data independence

Logical data independence indicates that the conceptual schema can be changed without affecting the existing external schemas. The change would be absorbed by the mapping between the external and conceptual levels.



Physical data independence

Physical data independence indicates that the physical storage structures or devices could be changed without affecting conceptual schema. The change would be absorbed by the mapping between the conceptual and internal levels.



Physical data independence

- If there is a need to change the file organization or the type of physical device used as a result of growth in the database or new technology, a change is required in the conceptual/internal mapping between the conceptual and internal levels.
- The physical data independence criterion requires that the conceptual level does not specify storage structures or the access methods (indexing, hashing etc.) used to retrieve the data from the physical storage medium.

Data Independence

- The Logical data independence is difficult to achieve than physical data independence, because it requires the flexibility in the design of database and programmer has to foresee the future requirements or modifications in the design.
- Applications program are more close to logic so logical data independence is difficult to achieve than physical data independence

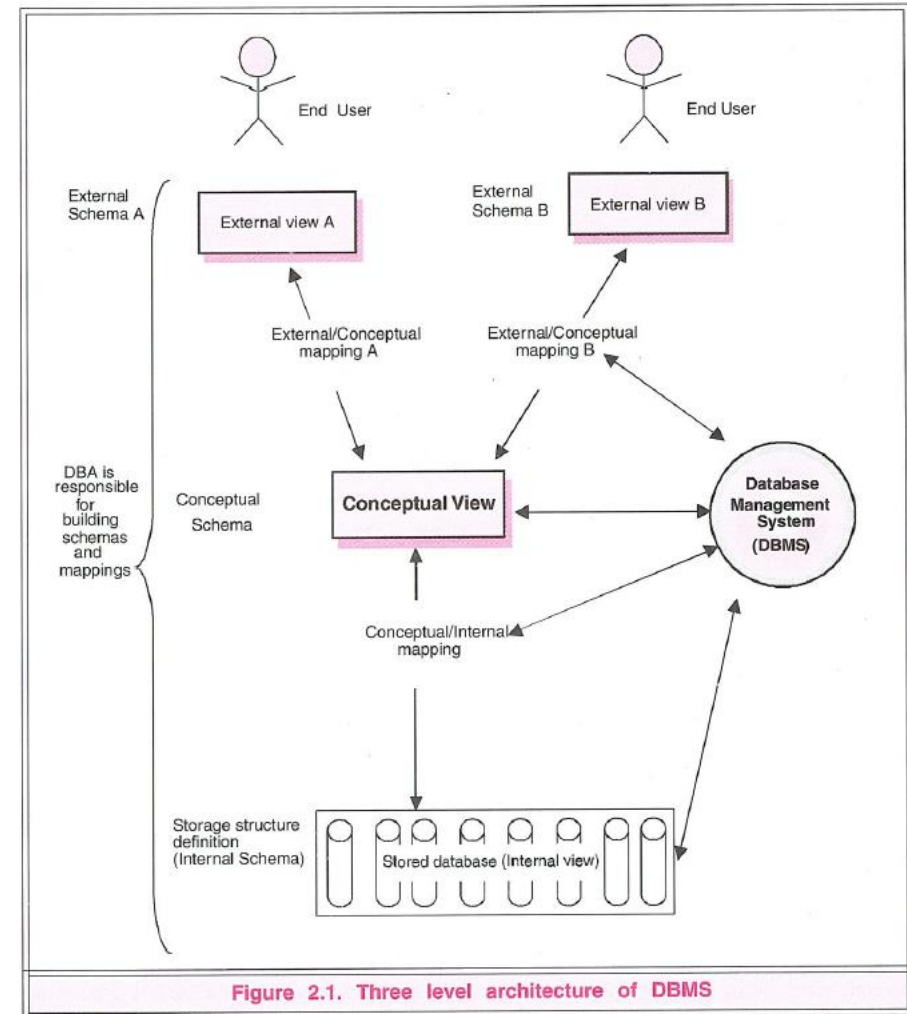
Three Level Architecture

Mapping between the Views

Mapping between Views

External/Conceptual Mapping:

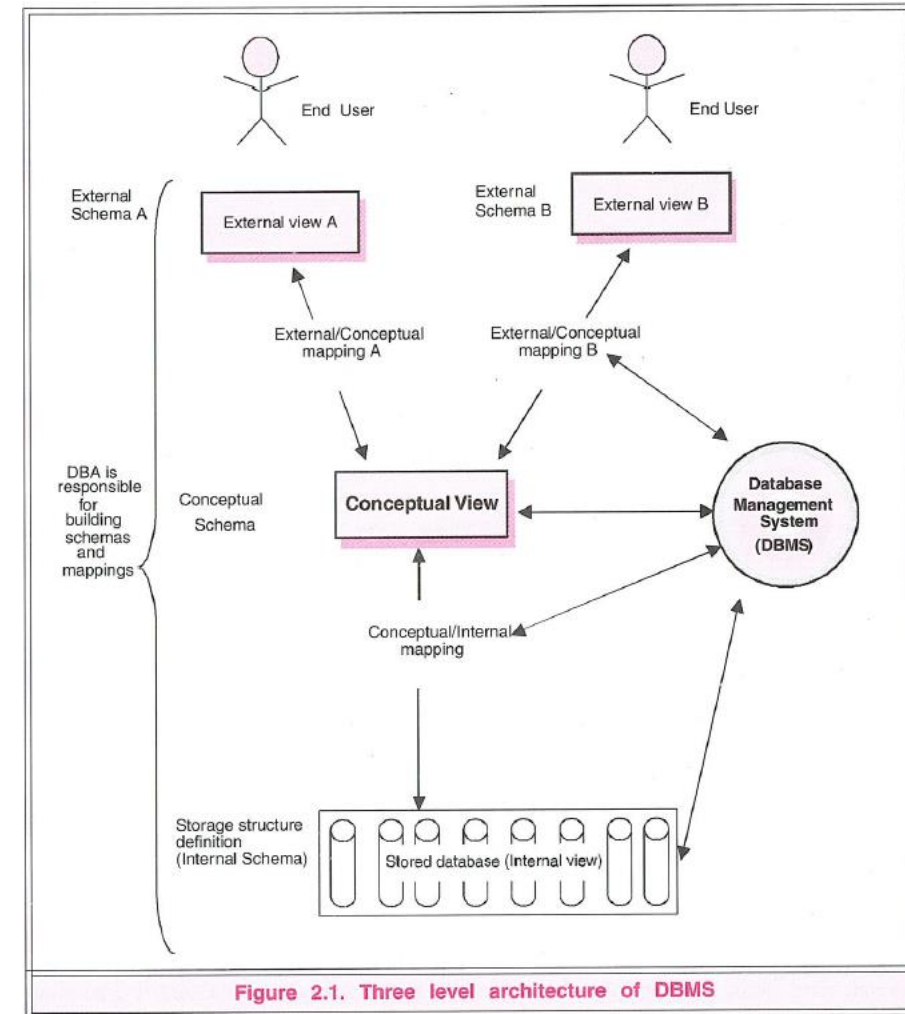
- Each external schema is related to the conceptual schema by the external/conceptual mapping.
- It maps logical record in the external view to one (or more) conceptual record(s) in the conceptual view.



Mapping between Views

External/Conceptual Mapping:

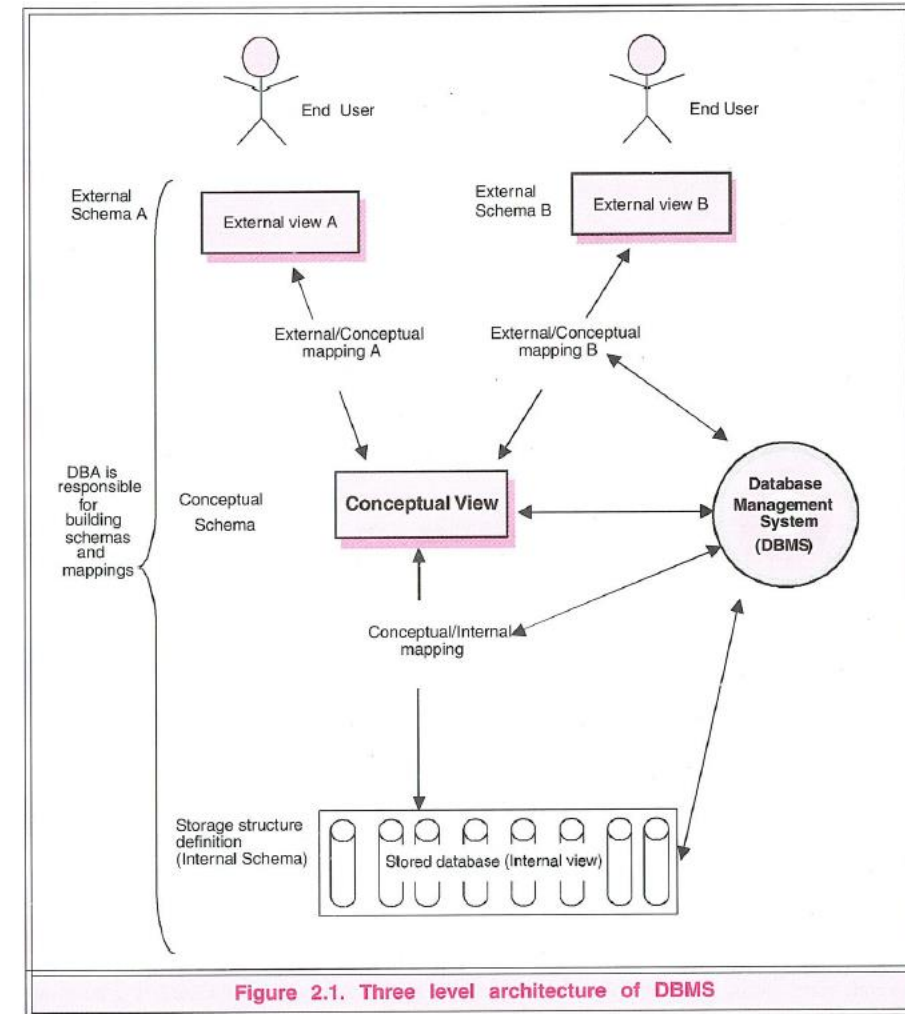
- A number of conceptual fields can be combined into a single external field, for example, Last_Name and First_Name at the conceptual level but Name at the external level.
- A given external record could be derived from a number of conceptual records.



Mapping between Views

Conceptual/Internal Mapping:

- Conceptual schema is related to the internal schema by the conceptual/internal mapping.
- This enables the DBMS to find the actual record or combination of records in physical storage that constitute a logical record in conceptual schema.



Data Independence

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Schema

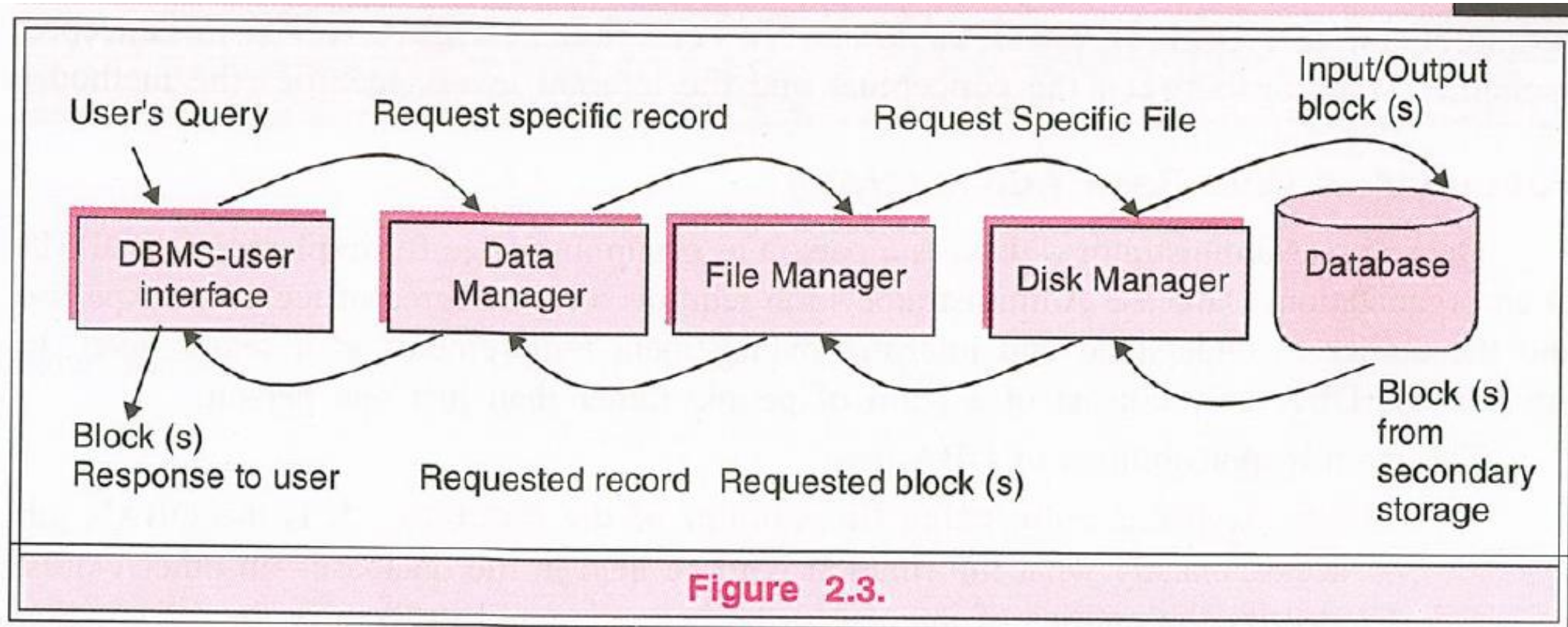
- **The overall description of the database is called the Database Schema.**
- There are three different types of schema in the database corresponding to each data view of database. In other words, the data views at each of three levels are described by schema.

Schema

- The External view is described by means of a schema called **external schema** that correspond to different views of the data.
 - Similarly the Conceptual view is defined by **conceptual schema**, which describes all the entities, attributes, and relationship together with integrity constraints.
 - Internal View is defined by **internal schema**, which is a complete description of the internal model, containing definition of stored records, the methods of representation, the data fields, and the indexes used.
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Procedure for Data Access by DBMS

Procedure for Data Access by DBMS

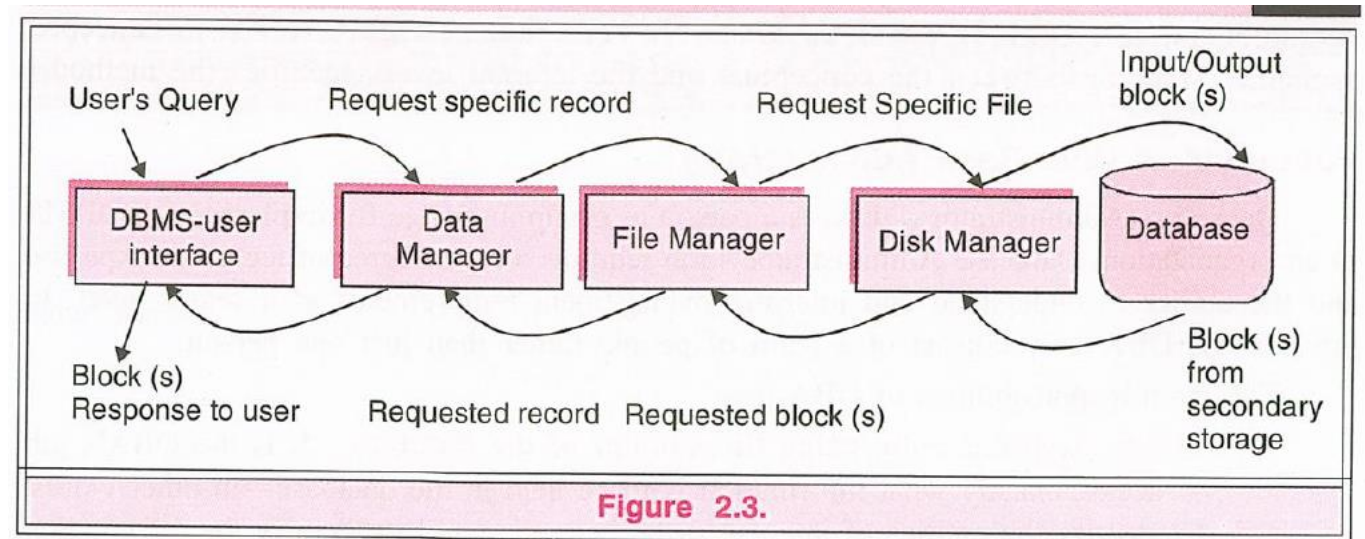


Procedure for Data Access by DBMS

Data Manager

A user's request for data is received by the data manager, which determines the physical record required.

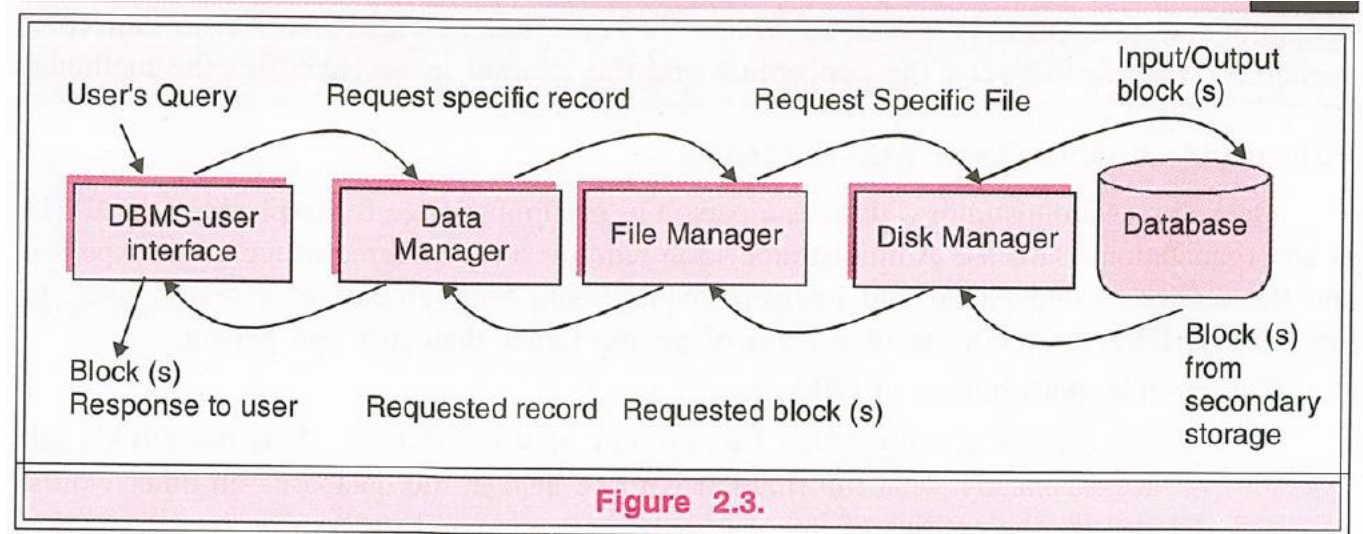
The data manager sends the request for a specific physical record to the file manager.



Procedure for Data Access by DBMS

DBMS User Interface

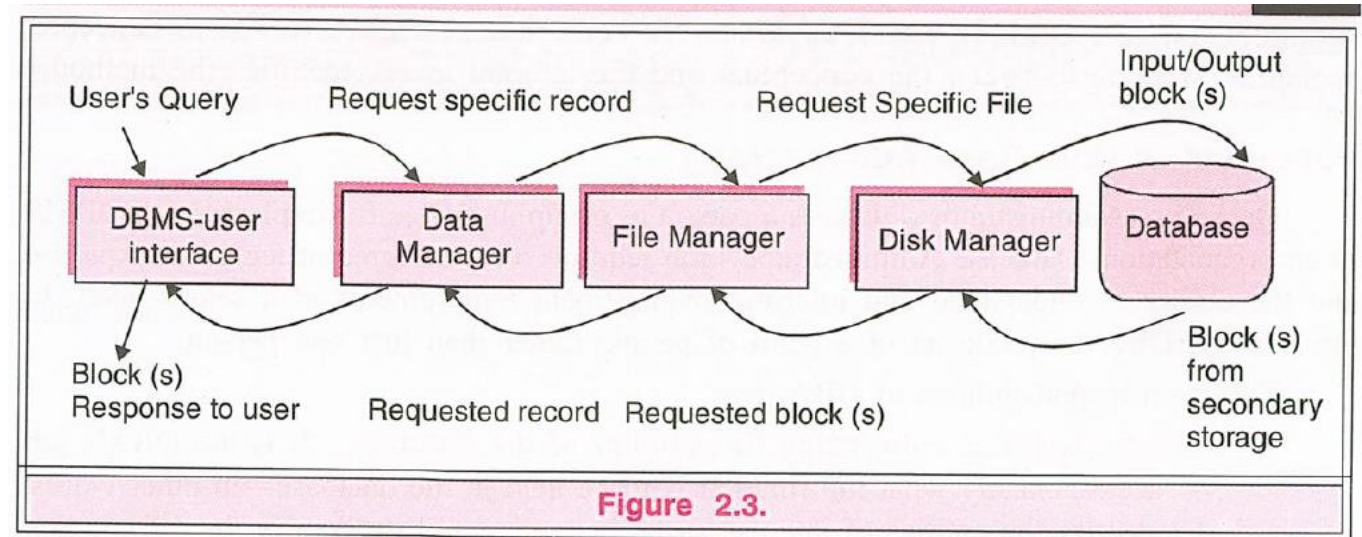
- It is used to query the database by the user and all the requests to access the desired data is given through this unterface.



Procedure for Data Access by DBMS

File Manager

- The file manager decides which physical block of secondary storage devices contains the required record and sends the request for the appropriate block to the disk manager.



Procedure for Data Access by DBMS

Disk Manager

- The disk manager retrieves the block and sends it to the file manager, which sends the required record to the data manager, which sends the required record to the DBMS-user interface, which sends the response to the user.
- A block is a unit of physical input/output operations between primary and secondary storage.

